

Contact

vireshduvvuri@gmail.com

www.linkedin.com/in/viresh-duvvuri
(LinkedIn)

viresh-duvvuri.netlify.app/
(Portfolio)

Top Skills

voice agents

guardrails

agent evaluation

Languages

English (Full Professional)

Hindi (Professional Working)

Telugu (Native or Bilingual)

Certifications

RAG system essentials

Introduction to Langchain

Supervised Machine Learning:
Regression and Classification

Coding essentials For AI Agents

Building a Multi-Agent System for
Software Engineering and Testing

Honors-Awards

1st Runner Up

Assistant Director

1st Runner Up - Best Poster

Publications

Development of Baton - A Low Cost
Aerial Delivery Vehicle

Viresh Duvvuri

Building Intelligent Software Solutions

United States

Summary

Building production software systems that solve real technical problems.

Currently developing AI agents and automation tools using LangChain, MCP, and Python that improve operational efficiency by 50-80%. Recent work includes multi-agent systems

for analytics and RAG-based diagnostic tools in production use.

Experience spans

full-stack development, embedded systems, and distributed applications across robotics and cloud platforms.

RECENT WORK:

- Multi-agent AI system for analytics (GridCOP) - reduced analyst workflows by 70%
- AI-powered diagnostic tool for log analysis - in production use
- Autonomous flight control systems (C++, PX4)
- Industrial robotics control systems (Python, ROS2)

EXPERIENCE:

5+ years developing production systems across robotics, AI applications, and distributed systems. Started in robotics and embedded systems (PX4, ROS2, autonomous drones), now focused on AI agent development and intelligent automation. Strong foundation in software engineering fundamentals with experience scaling from prototype to production.

Based in Seattle. Open to AI Engineer, Software Engineer, Robotics Engineer, and Backend

Engineer roles where I can build robust, scalable systems.

Let's connect if you're building software that solves real problems.

Experience

Grid CoOperator

AI Engineer

March 2025 - Present (1 year 2 months)

Seattle, WA

Grid Cop AI Agent Demo: <https://youtu.be/lbg38E8qwRk>

- Developed AI-enabled data processing system using LangChain, Python, and SQL databases from concept to deployment, reducing analyst research workflows by 70% within 2 months through intelligent query generation
- Built scalable backend service with API architecture handling 50-100 daily queries, ensuring reliable performance for real-time smart grid data analysis and operational decision support
- Implemented automated report generation pipeline accelerating stakeholder deliverables by 60% within first quarter, eliminating manual documentation processes for utility operations
- Designed context-aware response framework with dual validation system, improving data accuracy by 40% within 6 weeks through comprehensive error handling and quality assurance

Stealth Startup

AI Engineer - Voice

October 2025 - March 2026 (6 months)

Seattle, WA

Voice Ordering AI Agent Demo: <https://www.youtube.com/watch?v=TNmc9Z1LeKU>

- Architected and deployed production voice-based conversational AI agent serving as intelligent front desk automation, handling multi-turn dialogues for appointment scheduling, customer inquiries, and business workflow automation using MCP framework and advanced prompt engineering techniques

- Implemented comprehensive AI safety and monitoring infrastructure using OpenTelemetry to track conversation quality, detect hallucinations, and maintain response accuracy above 90% with real-time alerting and escalation protocols for production voice systems
- Built robust conversational AI pipelines with intent classification, entity extraction, and context management for business-specific scenarios, integrating with external APIs and knowledge bases to provide contextually appropriate responses across voice interactions
- Designed and deployed guardrails and fallback mechanisms to prevent AI hallucinations and ensure reliable handoff protocols when confidence thresholds are not met, maintaining high user satisfaction and system reliability for customer-facing voice applications

Freefly Systems

4 years

Software Engineer (Forward Deployed)

November 2023 - October 2025 (2 years)

Drone Log Analysis AI Agent Demo: https://youtu.be/0mGi4_23IrY

- Developed comprehensive diagnostic and analysis tools for engineering teams, building automated systems to process complex technical data and identify system failures
- Independently designed and built AI-powered diagnostic tool using Python and modern LLM frameworks from requirements to production
- Built knowledge base enhancements and automated support tools, improving customer self-service capabilities and team response times by 40%

Software Engineer

November 2021 - October 2023 (2 years)

Woodinville, WA

- Contributed to drone platform codebases, implementing new features and optimizations for flight control systems and payload integration across multiple product lines

- Managed software integration projects from planning through release, coordinating firmware updates and executing comprehensive testing protocols with cross-functional teams
- Led release management for drone platforms, overseeing testing phases from alpha through production deployment while maintaining communication with engineering and management teams

Lumenier

Software Engineer

July 2020 - November 2021 (1 year 5 months)

Sarasota, FL

Room Surveillance Simulation Demo: https://youtu.be/SHDK3_jQy_o

Toss to launch - custom flight mode - Demo: <https://youtu.be/8Z5y0LJhIJo>

- Implemented custom software using C++ and data structures for specialized applications, enabling autonomous capabilities within 8 weeks
- Enhanced system performance through algorithms and data ingestion pipelines, improving operational efficiency by 45% across environments
- Architected testing frameworks with software design principles, reducing implementation issues by 30% within 3 months

York Exponential

Software Engineer

August 2018 - March 2020 (1 year 8 months)

York, PA

Weld bot HMI (Human Machine Interface) Demo: <https://youtu.be/GWhB5x7PyIE>

- Created Human Machine Interface for collaborative welding using Python and ROS2, reducing operator complexity by 50% within 4 months
- Developed autonomous robot prototype using computer vision and machine learning from requirements to working deployment
- Implemented features for a Multi-Robot Control System, focusing on platform independence and system reliability.

Washington State University
Post Masters - Research Assistant
October 2017 - August 2018 (11 months)
Pullman, WA, USA

Baton Drone - Drop and recovery - Functionality test simulation Demo: <https://youtu.be/SCrroi1DFfE>

- Led the development of a precision delivery drone for military and rescue operations, integrating critical payload systems.
- Integrated Px4-based flight controller with autonomous air-drop, hover, and precision landing capabilities.

Washington State University
1 year 8 months
Research Assistant
December 2015 - July 2017 (1 year 8 months)
Pullman, USA

- Implemented control equations and programmed flight controller for Baton Drone project.
- Analyzed card swipe data and developed web application for Student Flow Analysis project.
- Presented research posters at prestigious symposiums and expos.

Teaching Assistant
August 2016 - December 2016 (5 months)
Pullman, WA

- Assisted undergraduate students at Washington State University with course-related queries and problems.
- Designed and evaluated challenging problems for course examinations to enhance student learning outcomes.

Washington State University Office of Commercialization
Technology Transfer Intern
May 2016 - August 2016 (4 months)
Pullman, WA

- Assessed and evaluated technologies for commercialization purposes at Washington State University Office of Commercialization.
- Wrote non-confidential reports detailing commercialization potential and identified market players in the space.

- Analyzed the competitive landscape to support strategic decision-making and enhance market positioning.

Education

Washington State University

Master's degree, Computer Science · (August 2015 - July 2017)

GITAM UNIVERSITY

Bachelor of Technology (BTech), Information Technology · (2011 - 2015)